

Data 236- Distributed Systems

Homework-3

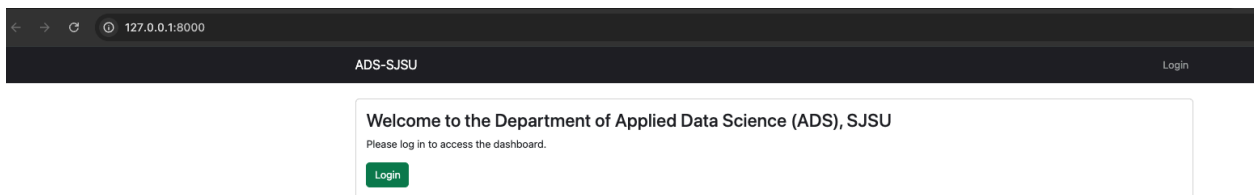
Kruthika Virupakshappa

Part 1-FastAPI Application

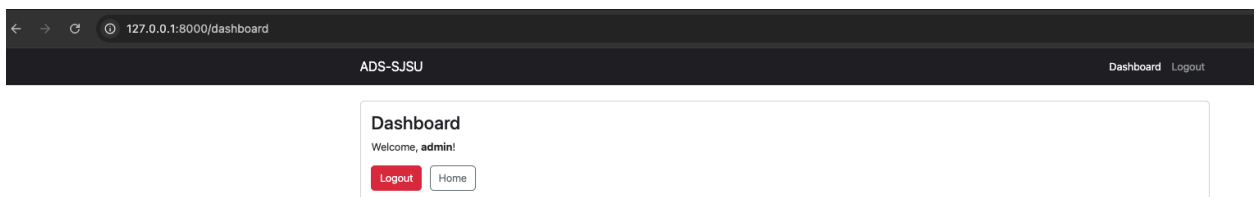
1. Routes - Handled Separately in a router:

- **Home Page (/):** Display a welcome message for ADS-SJSU. Show a link to the login page if the user is not logged in. Show a link to the dashboard and log out if the user is logged in.

Screenshot: Login link when user is not logged in

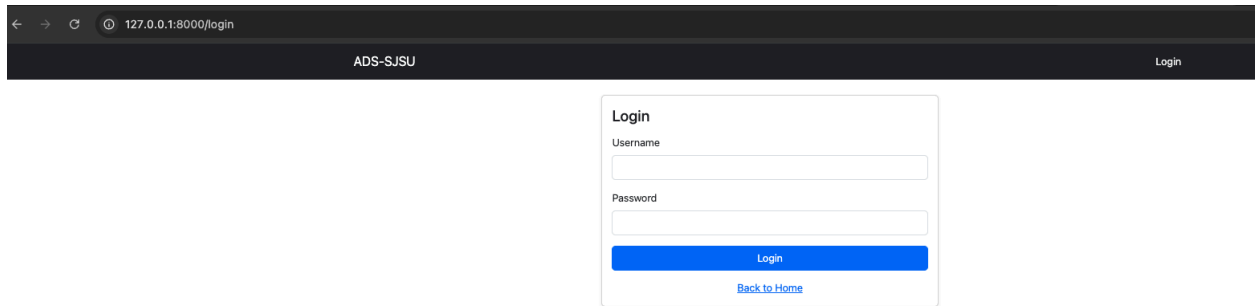


- Screenshot: Dashboard link when user is logged in

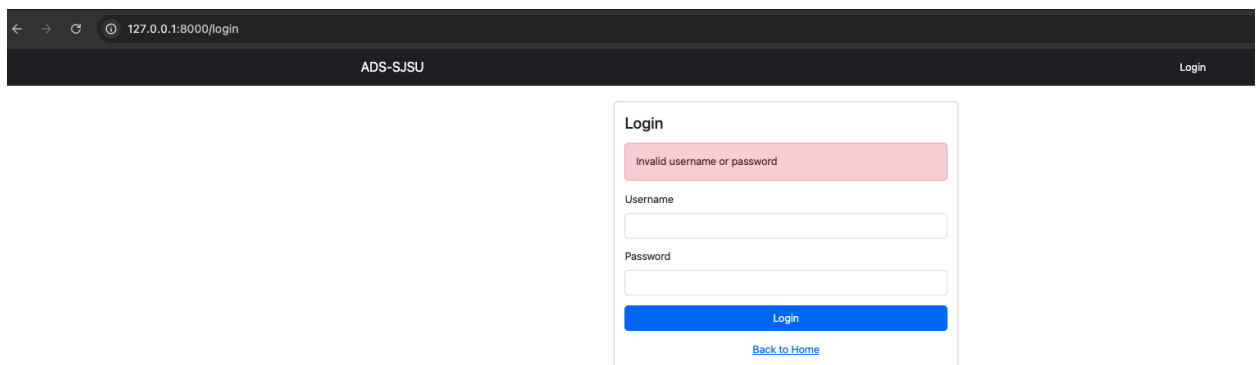


- **Login Page (/login):** Display a login form with fields for username and password. Validate the credentials and log the user in if they are correct. Redirect to the dashboard on successful login. Display a Bootstrap alert message on the login page when a user enters invalid credentials.

Screenshot: Login form with username and password

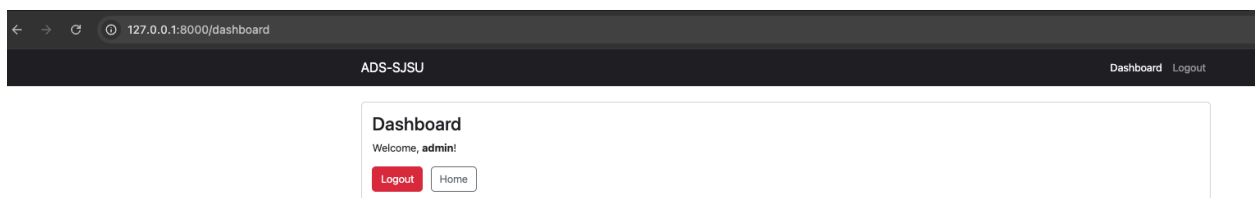


Screenshot: Invalid username/password:



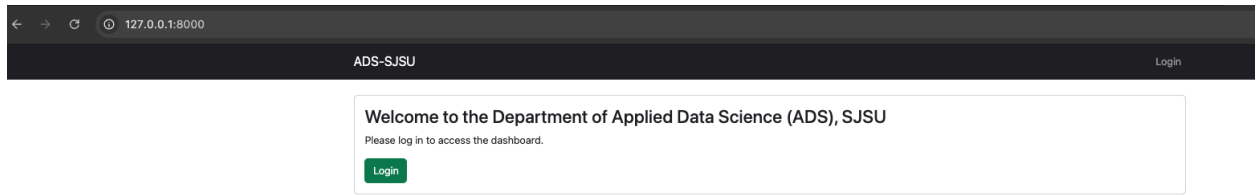
- **Dashboard Page (/dashboard):** Display a welcome message with the user's name. Show a logout link. Protect this route so only logged-in users can access it.

Screenshot: Welcome message



- **Logout (/logout):** Destroy the session and redirect the user to the home page.

Screenshot: User redirected to home page after logout



2. Session Management: Use Starlette's SessionMiddleware to manage user sessions. Store the logged-in user's information in the session. Ensure that the session cookie is secure.

main.py

```
import os
import uvicorn
from fastapi import FastAPI
from starlette.middleware.sessions import SessionMiddleware
from routers.auth import router as auth_router

# Create FastAPI app
app = FastAPI(title="ADS-SJSU Authentication System")

# Secret key for session signing
SECRET_KEY = os.getenv("SECRET_KEY", "dev-only-secret-key")

# Enable session support
app.add_middleware(
    SessionMiddleware,
    secret_key=SECRET_KEY,
    https_only=True,
    same_site="lax",
    max_age=3600
)

# Register routes
app.include_router(auth_router)

if __name__ == "__main__":
    uvicorn.run(
        "main:app",
        host="127.0.0.1",
        port=8000,
        reload=True
    )
```

3. Styling with Bootstrap: Explore and use Bootstrap to style all pages. Make the application responsive and visually appealing. Use Bootstrap components, such as the Navbar for navigation, Cards for forms and content, Buttons for actions, and Alerts for messages.

Include one screenshot per route page. Include only the auth.py code at the bottom and one screenshot of your templates directory.

auth.py

```
from fastapi import APIRouter, Request, Form
from fastapi.responses import RedirectResponse
from fastapi.templating import Jinja2Templates
from starlette.status import HTTP_302_FOUND

router=APIRouter()

templates = Jinja2Templates(directory="templates")

VALID_USERNAME = "admin"
VALID_PASSWORD = "password"

@router.get("/")
def home(request:Request):
    user = request.session.get("user")
    return templates.TemplateResponse(
        "index.html",
        {
            "request": request,
            "user":user
        }
    )

@router.get("/login")
def login_page(request: Request):
    user = request.session.get("user")

    return templates.TemplateResponse(
        "login.html",
        {
            "request": request,
            "user": user
        }
    )

@router.post("/login")
def login(request: Request, username: str = Form(...), password: str = Form(...)):
    """
    Handles login form submission.

    - Reads username and password from the form
    - Validates credentials
    - Stores user info in session if valid
    """
```

```

- Displays Bootstrap alert if invalid
"""
if username == VALID_USERNAME and password == VALID_PASSWORD:
    # Store logged-in user in session
    request.session["user"] = username

    # Redirect user to dashboard
    return HttpResponseRedirect(
        url="/dashboard",
        status_code=HTTP_302_FOUND
    )
    # If credentials are invalid:
# Re-render login page with error message
return templates.TemplateResponse(
    "login.html",
    {
        "request": request,
        "user": None,
        "error": "Invalid username or password"
    }
)

@router.get("/dashboard")
def dashboard(request: Request):
    """
    Protected route.

    - Only accessible if user is logged in
    - Redirects to login page if session is missing
    """
    user = request.session.get("user")

    # If user is not logged in, block access
    if not user:
        return HttpResponseRedirect(
            url="/login",
            status_code=HTTP_302_FOUND
        )

    # If user is logged in, render dashboard
    return templates.TemplateResponse(
        "dashboard.html",
        {
            "request": request,
            "user": user
        }
    )

@router.get("/logout")
def logout(request: Request):

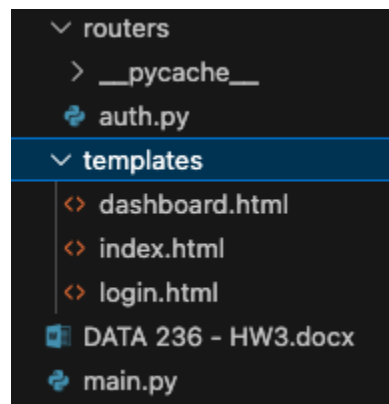
```

```
"""
Logs the user out.

- Clears all session data
- Redirects back to home page
"""
request.session.clear()

return RedirectResponse(
    url="/",
    status_code=HTTP_302_FOUND
)
```

Template Directory:



Login.html

```
templates > login.html > html > body > div.container.mt-4
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1" />
6      <title>ADS-SJSU Auth App - Login</title>
7      <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
8  </head>
9  <body>
10
11  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
12      <div class="container">
13          <a class="navbar-brand" href="/">ADS-SJSU</a>
14          <div class="navbar-nav">
15              {% if user %}
16                  <a class="nav-link" href="/dashboard">Dashboard</a>
17                  <a class="nav-link" href="/logout">Logout</a>
18              {% else %}
19                  <a class="nav-link active" href="/login">Login</a>
20              {% endif %}
21          </div>
22      </div>
23  </nav>
24
25  <div class="container mt-4">
26      <div class="row justify-content-center">
27          <div class="col-md-5">
28              <div class="card shadow-sm">
29                  <div class="card-body">
30                      <h4 class="card-title mb-3">Login</h4>
31
32                      {% if error %}
33                          <div class="alert alert-danger role="alert">
34                              {{ error }}
35                          </div>
36                      {% endif %}
37
38                      <form method="post" action="/login">
39                          <div class="mb-3">
40                              <label class="form-label">Username</label>
41                              <input class="form-control" type="text" name="username" required>
42                          </div>
43
44                          <div class="mb-3">
45                              <label class="form-label">Password</label>
46                              <input class="form-control" type="password" name="password" required>
47                          </div>
48
49                          <button class="btn btn-primary w-100" type="submit">Login</button>
50                      </form>
51
52                      <div class="text-center mt-3">
53                          <a href="/">Back to Home</a>
54                      </div>
55                  </div>
56              </div>
57          </div>
58      </div>
59  </div>
60 </body>
61 </html>
```

Index.html

```
templates > index.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1" />
6      <title>ADS-SJSU Auth App</title>
7      <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
8  </head>
9  <body>
10
11  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
12      <div class="container">
13          <a class="navbar-brand" href="/">ADS-SJSU</a>
14          <div class="navbar-nav">
15              {% if user %}
16                  <a class="nav-link" href="/dashboard">Dashboard</a>
17                  <a class="nav-link" href="/logout">Logout</a>
18              {% else %}
19                  <a class="nav-link" href="/login">Login</a>
20              {% endif %}
21          </div>
22      </div>
23  </nav>
24
25  <div class="container mt-4">
26      <div class="card">
27          <div class="card-body">
28              <h3 class="card-title">Welcome to the Department of Applied Data Science (ADS), SJSU</h3>
29
30              {% if user %}
31                  <p class="card-text">You are logged in as <strong>{{ user }}</strong>.</p>
32                  <a class="btn btn-primary" href="/dashboard">Go to Dashboard</a>
33                  <a class="btn btn-outline-danger ms-2" href="/logout">Logout</a>
34              {% else %}
35                  <p class="card-text">Please log in to access the dashboard.</p>
36                  <a class="btn btn-success" href="/login">Login</a>
37              {% endif %}
38          </div>
39      </div>
40  </div>
41
42  </body>
43  </html>
44
```

Dashboard.html

templates > dashboard.html > ...

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  |   <meta charset="UTF-8" />
5  |   <meta name="viewport" content="width=device-width, initial-scale=1" />
6  |   <title>ADS-SJSU Auth App - Dashboard</title>
7  |   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
8  </head>
9  <body>
10
11  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
12  |   <div class="container">
13  |     <a class="navbar-brand" href="/">ADS-SJSU</a>
14  |     <div class="navbar-nav">
15  |       <a class="nav-link active" href="/dashboard">Dashboard</a>
16  |       <a class="nav-link" href="/logout">Logout</a>
17  |     </div>
18  |   </div>
19  </nav>
20
21  <div class="container mt-4">
22  |   <div class="card">
23  |     <div class="card-body">
24  |       <h3 class="card-title">Dashboard</h3>
25  |       <p class="card-text">Welcome, <strong>{{ user }}</strong>!</p>
26  |
27  |       <a class="btn btn-danger" href="/logout">Logout</a>
28  |       <a class="btn btn-outline-secondary ms-2" href="/">Home</a>
29  |     </div>
30  |   </div>
31  </div>
32
33  </body>
34  </html>
35
```